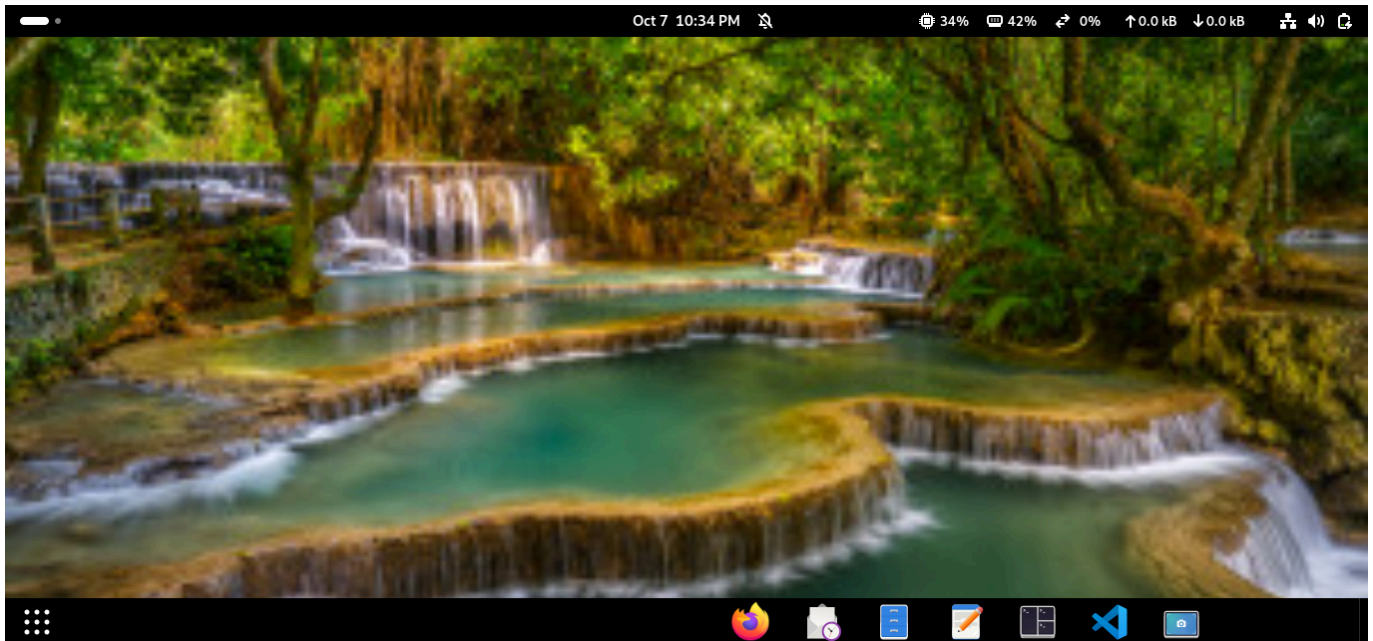
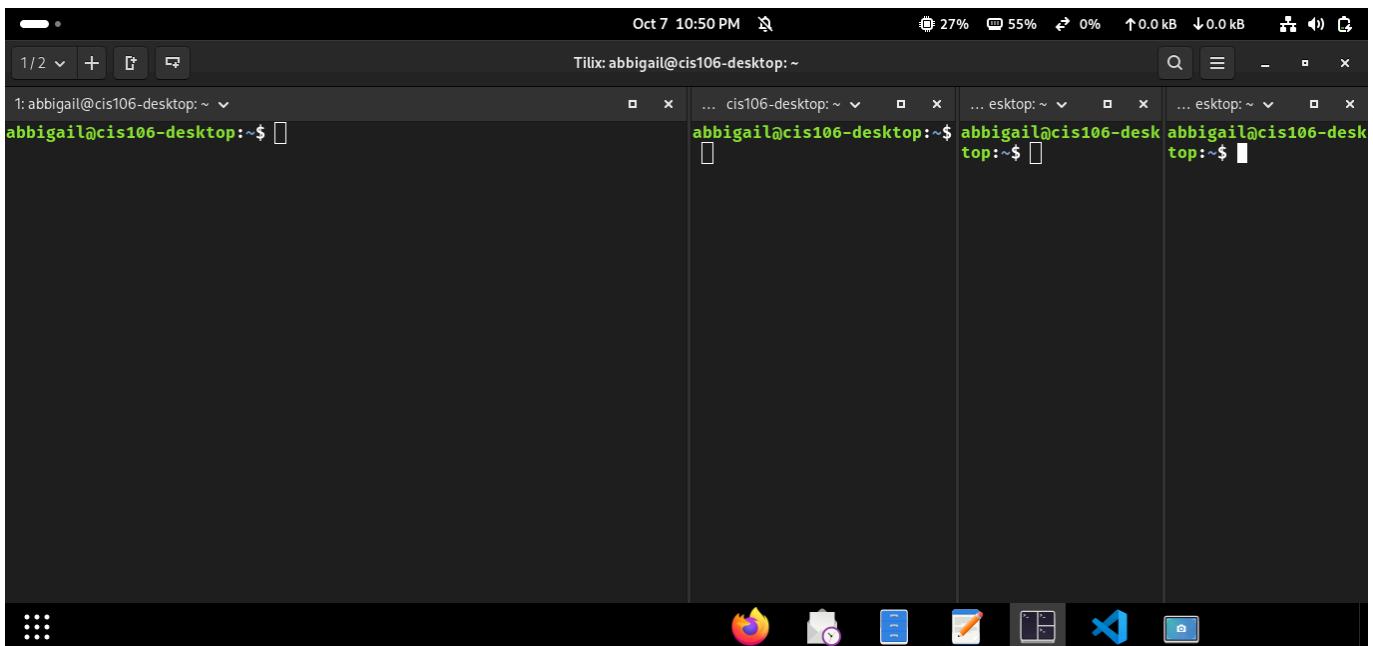


Lab 3 Submission

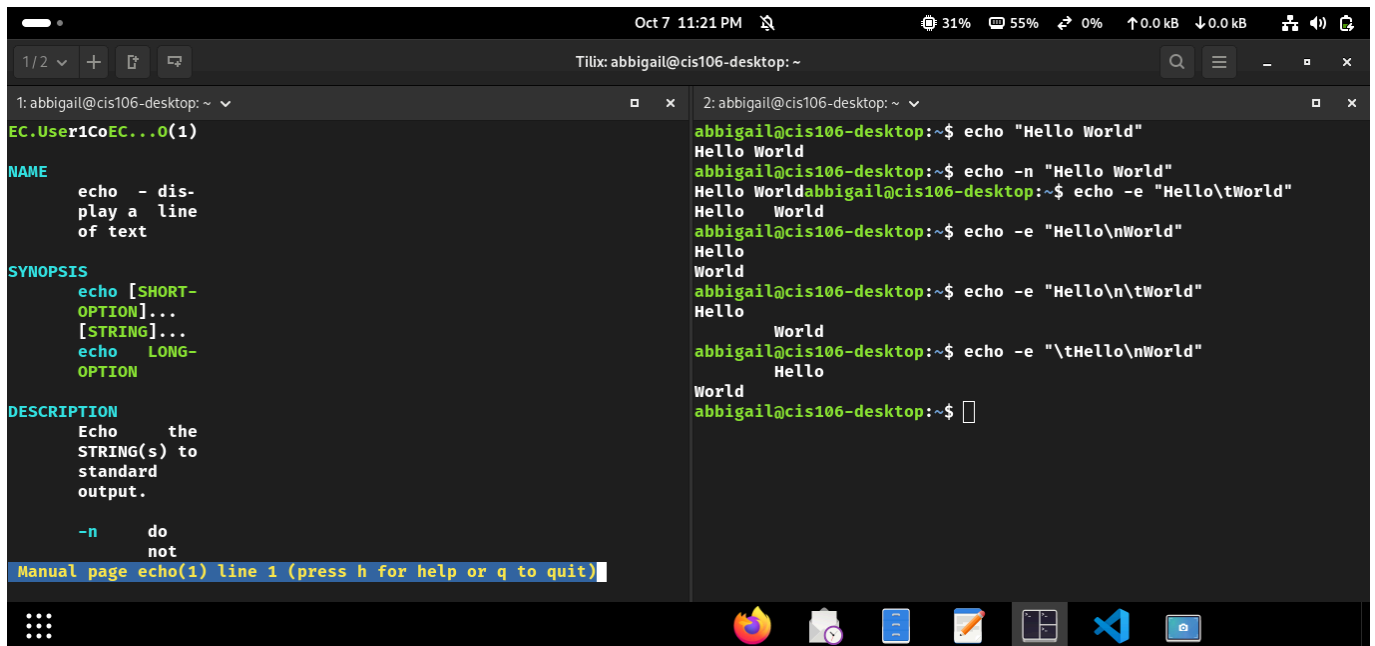
Question 1



Question 2



Question 3



```
Oct 7 11:21 PM 31% 55% 0% ↑ 0.0 kB ↓ 0.0 kB
1/2 + [ ] [ ]
Tilix: abbigail@cis106-desktop: ~
1: abbigail@cis106-desktop: ~
EC.User1CoEC...0(1)
NAME
    echo - display a line of text
SYNOPSIS
    echo [SHORT-OPTION]... [STRING]...
    echo LONG-OPTION
DESCRIPTION
    Echo the STRING(s) to standard output.
    -n do not
Manual page echo(1) line 1 (press h for help or q to quit)
2: abbigail@cis106-desktop: ~
abbigail@cis106-desktop:~$ echo "Hello World"
Hello World
abbigail@cis106-desktop:~$ echo -n "Hello World"
Hello Worldabbigail@cis106-desktop:~$ echo -e "Hello\tWorld"
Hello World
abbigail@cis106-desktop:~$ echo -e "Hello\nWorld"
Hello
World
abbigail@cis106-desktop:~$ echo -e "Hello\n\tWorld"
Hello
World
abbigail@cis106-desktop:~$ echo -e "\tHello\nWorld"
World
Hello
abbigail@cis106-desktop:~$
```

Challenge Question

1. John wants to see how much free memory he has in his system. He understands that the free command is used for this however, he would like the output of the command to appear in a single line and with human readable format.

What is the complete command that John has to use here? free -h -L

Take a screenshot showing the man page showing the correct options and the command and its output. See the example for a reference.

The image consists of two screenshots of a terminal window. The top screenshot shows the man page for the 'free' command, specifically the '-h, --human' option. The bottom screenshot shows the same terminal window with the 'free -h -L' command executed, displaying memory usage statistics.

Top Screenshot: Man page for free(1)

```
1: abbigail@cis106-desktop: ~  
-h, --human  
    Show all output  
    fields automati-  
    cally scaled to  
    shortest three  
    digit unit and  
    display the units  
    of print out.  
    Following units  
    are used.  
  
    B = bytes  
    Ki = kibibyte  
    Mi = mebibyte  
    Gi = gibibyte  
    Ti = tebibyte  
    Pi = pebibyte  
  
    If unit is miss-  
    ing, and you have  
    exbibyte of RAM or  
Manual page free(1) line 145 (press h for help or q to quit)
```

Bottom Screenshot: free command output

```
2: abbigail@cis106-desktop: ~  
abbigail@cis106-desktop:~$ free -h -L  
SwapUse 524Ki CachUse 799Mi MemUse 1.6Gi MemFree  
679Mi  
abbigail@cis106-desktop:~$
```

2. Kenny is a system administrator working on automation scripts. He is building a script where he needs to know the kernel name, kernel release, kernel version, and operating system of the computer. He knows that the `uname` command can provide this information.

What is the complete command that Kenny has to use here? `uname -s -r -v -o`

Take a screenshot showing the man page showing the correct options and the command and its output. See the example for a reference.

```

1: abbigail@cis106-desktop: ~
    print the proces-
    sor type
    (non-portable)

-i, --hardware-platform
    print the hardware
    platform
    (non-portable)

-o, --operating-system
    print the operat-
    ing system

--help display this help
    and exit

--version
    output version in-
    formation and exit

AUTHOR
    Written by David MacKen-
    zie.

2: abbigail@cis106-desktop: ~
    abbigail@cis106-desktop:~$ free -h -L
    SwapUse 524Ki CachUse 799Mi MemUse 1.6Gi MemFree
    679Mi
    abbigail@cis106-desktop:~$ uname -s -r -v -o
    Linux 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1
    (2026-08-29) GNU/Linux
    abbigail@cis106-desktop:~$
  
```

```

1: abbigail@cis106-desktop: ~
    -s, --kernel-name
    print the kernel
    name

-n, --nodename
    print the network
    node hostname

-r, --kernel-release
    print the kernel
    release

-v, --kernel-version
    print the kernel
    version

-m, --machine
    print the machine
    hardware name

-p, --processor
    print the proces-
    sor type
    (non-portable)

2: abbigail@cis106-desktop: ~
    abbigail@cis106-desktop:~$ free -h -L
    SwapUse 524Ki CachUse 799Mi MemUse 1.6Gi MemFree
    679Mi
    abbigail@cis106-desktop:~$ uname -s -r -v -o
    Linux 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1
    (2026-08-29) GNU/Linux
    abbigail@cis106-desktop:~$
  
```

- Dave is working with an API that only accepts rfc-3339 time format for dates. For example, 2025-09-10 18:39:53.467197335-04:00 is an acceptable format but anything else will return an error. He needs a date command that will allow him to convert the current time into rfc-3339 format.

What is the complete command that Dave has to use here? `date --rfc-3339=ns`

Take a screenshot showing the man page showing the correct options and the command and its output. See the example for a reference.

The screenshot shows a terminal window with two panes. The left pane displays the help text for the 'date' command, and the right pane shows the output of system-related commands.

```
Oct 7 11:44 PM 24% 56% 0% ↑ 0.0 kB ↓ 0.0 kB
```

1: abbigail@cis106-desktop: ~

```
output the available resolution of timestamps Example: 0.000000001

-R, --rfc-email
output date and time in RFC 5322 format. Example:
Mon, 14 Aug 2006 02:34:56 -0600

--rfc-3339=FMT
output date/time in RFC 3339 format. FMT='date',
'seconds', or 'ns' for date and time to the indicated
precision. Example: 2006-08-14 02:34:56-06:00

-r, --reference=FILE
display the last modification time of FILE

-s, --set=STRING
set time described by STRING

-u, --utc, --universal
print or set Coordinated Universal Time (UTC)

--help display this help and exit
Manual page date(1) line 34/235 29% (press h for help or q to quit)
```

2: abbigail@cis106-desktop: ~

```
abbigail@cis106-desktop:~$ free -h -L
SwapUse      524Ki CachUse      799Mi MemUse      1.6Gi MemFree
679Mi
abbigail@cis106-desktop:~$ uname -s -r -v -o
Linux 6.12.107+deb13-amd64 #1 SMP PREEMPT_DYNAMIC Debian 6.12.107-1
abbigail@cis106-desktop:~$ date --rfc-3339=ns
2026-10-07 23:44:01.927881470-04:00
abbigail@cis106-desktop:~$
```

5 / 5